

THREE-DIMENSIONAL SOFT TISSUE PRESERVATION REVEALED IN THE SKIN OF A NON-AVIAN DINOSAUR

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Typescript received 2 October 2018; accepted in revised form 19 November 2019

Abstract: The most commonly preserved soft tissues associated with ornithischian dinosaurs are skin remains. The apparent resistance of hadrosaur skin to decay, and its abundance in the fossil record relative to that of other tetrapods, has been attributed to factors such as thickness and composition. Here we report additional intrinsic factors within hadrosaur skin: 3D-preserved eumelanin-bearing bodies,

dermal cells and blood vessel fragments in an organic matrix composed of protein fossilization products. The skin is much thinner than that of living mammals of similar size. It is likely that the preservation of hadrosaur skin is related to the arrangement of the layers composing it.

Key words: skin, taphonomy, hadrosaur, melanin, soft tissue.

THE skin is the largest organ in the body of any vertebrate and carries out important functions such as homeostasis, sensory reception and visual signaling (Landmann 1986). The skin is composed of three layers: the epidermis (outer), dermis and the subcutis (innermost). The epidermis is composed of stratified layers of keratinized cells (the stratum corneum, stratum intermedium and stratum basale) whereas the dermis is characterized by a stratum spongiosum of loose collagen fibres and a lower stratum compactum of tightly packed, orthogonal arrays of collagen fibres (Landmann 1986; Cooper & Greenberg 1992). Colouration of the skin is generally determined by melanophores and chromatophores and the structural layering of the integument (McNamara *et al.* 2016; Shawkey & D’Alba 2017). Chromatophores and melanophores are usually stored at the interface between the epidermis and dermis in vertebrates (Prum & Torres 2003, 2004; Chang *et al.* 2009; McNamara *et al.* 2016; Shawkey & D’Alba 2017). The innermost layer of the skin, the subcutaneous layer, separates the integument from the rest of the body. Integument is probably the most abundant type of soft tissue in the vertebrate fossil record (Schweitzer 2011; Davis 2014) and it is providing increasingly important evidence of reptile evolution and ecological diversity in the past (e.g. Martill 1995; Briggs *et al.* 1997; Martill *et al.* 2000; Bell 2012; Lindgren *et al.* 2013, 2014, 2018; Davis 2014; McNamara *et al.* 2016, 2018; Yang *et al.* 2019).

Fossilized skin is preserved in three main ways: as impressions, compressions or permineralized, three-dimensional skin (Schweitzer 2011). Impressions are occasionally found in association with bones or footprints (e.g. Lockley *et al.* 2004; Paik *et al.* 2017). Compressions are preserved as a carbon-rich layer surrounding bones (e.g. Briggs *et al.* 1997; Qiang *et al.* 1998; Wang *et al.* 2019). Permineralized skin is the rarest of the three modes of preservation (Schweitzer 2011; Bell 2012; Davis 2014) and usually occurs in association with three dimensionally preserved skeletons. Permineralized skin was previously thought to be a high-fidelity mineralized replication of the original organic tissue (Schweitzer 2011). However, recent investigations of a few exceptionally preserved examples have challenged this view by recovering original organic molecules. Dinosaurian integument preserved as a result of early mineralization is known in Ornithischia, Sauropoda and Theropoda (Briggs *et al.* 1997; Martill *et al.* 2000; Bell 2012; Davis 2014; McNamara *et al.* 2018). The preservation of integument is most prevalent in hadrosaurs (Bell 2012; Davis 2014). However, the taphonomic processes that favour the fossilization of skin and associated molecular preservation are still poorly understood (Lyson & Longrich 2011; Bell 2012; Davis 2014). Previous investigations of dinosaur skin were performed on the basal ceratopsian *Psittacosaurus* (Lingham-Soliar 2008), a nodosaur (Brown

et al. 2017), and coelurosaurs (Navalón *et al.* 2015; Bell *et al.* 2017; McNamara *et al.* 2018). Traces of collagenous material were found in a ‘mummified’ hadrosaurid from North Dakota, USA (Manning *et al.* 2009). Skin colouration patterns have been reconstructed in ornithischian dinosaurs (Lingham-Soliar & Plodowski 2010; Vinther *et al.* 2016; Brown *et al.* 2017) but not in a hadrosaurid.

YPMPU 016969 is a partial hadrosaur skeleton preserving skin in the flank region of the body (Fabbri *et al.* 2019, fig. S1). We investigated this specimen with the following aims: (1) to understand its taphonomy; (2) to test for soft tissue preservation and the presence of melanin; and (3) to consider previous explanations for the prevalence of hadrosaur skin preservation in the fossil record.

MATERIAL AND METHOD

YPMPU 016969 was selected for its combination of 3D preservation of the integument (Fabbri *et al.* 2019, figs S1–S2), the presence of bone (fragments of the distal portion of a dorsal rib below the skin) and the lack of preparation (the skin is still largely covered by sediment and no consolidant has been applied) (specimen repository: Yale Peabody Museum of Natural History). We performed chemical analyses on the skin fragments and investigated the microstructure and nature of the preservation in a thin section (70 µm thick) prepared by standard methods (Chinsamy & Raath 1992) and left uncovered. The skin microstructure was investigated with a petrographic microscope Leica DM 2500 P and photographed with a ProgRes Cfsan camera. The elemental composition of the skin was determined by energy-dispersive x-ray spectroscopy (EDS) spot analysis and verified with EDS chemical mapping and Raman microspectroscopy. EDS analysis and SEM imaging were performed using a JXA-8530F scanning electron microscope.

EDS chemical mapping does not distinguish between organic and inorganic carbon. We therefore decalcified samples of YPMPU 016969 and applied Raman spectroscopy to investigate the molecular composition of the residues. Samples of epidermis, dermis, bone and inner and outer sediment were identified on the basis of observations on the thin sections and were obtained using a 0.2 mm blade rock saw, and decalcified by incubating 3 g in 2 ml Eppendorf tubes with 1.5 ml of a 0.1 M aqueous solution of hydrochloric acid (Merck, Germany) until complete structural disintegration was achieved. The residues were examined with a standard light microscope (Leica DM 2500 P). Raman microspectroscopy was performed immediately after decalcification using a Horiba

Jobin-Yvon HR800 with 532 nm excitation (20 mW at the sample surface). The scattered Raman light was detected by an electron multiplier charged-coupled device (EM-CCD) after being dispersed by an 1800 grooves/mm grating and passed through a 100 µm entrance slit (hole size 300 µm). The spectrometer was calibrated using the first order Si band at 520.7/cm. Six spectra were accumulated in the 500–1800/cm region for 20 s exposure time each. Spectra were processed with the software LabSpec5 and Spectragryph (Menges 2017). A standard automated spectral peak search was applied to the raw spectra in the SpectraGryph 1.2 software to eliminate background noise and select only significant peaks. Assignment of the peaks was based on Wiemann *et al.* (2018) for organic compounds and Huang *et al.* (2004) and Galván *et al.* (2013) for melanin (see Fabbri *et al.* 2019, table 1). To exclude the possibility of contamination, a dataset of Raman spectra of extant bone, skin tissues, fossils containing proteinaceous compounds and ‘blanks’ (i.e. glues, consolidant, fossils lacking organic components) was assembled from Wiemann *et al.* (2018) (for samples included in the analyses, see Fabbri *et al.* 2019, tables 2–3). We tested for the presence of contamination in our samples with principal component analysis (PCA) and cluster analysis. Because our investigation considered organic compounds only, we excluded peaks that are affected by the presence of clay minerals, such as 510, 630, 803, 830, 880 and 980 (Wiemann *et al.* 2018). We excluded eggshell material, because those organic compounds are very different from those found in other hard tissues such as teeth and bones (Wiemann *et al.* 2018). PCA was performed on the resulting spectra. Cluster and K-clustering analyses were run to identify associations based on spectra obtained from Raman spectroscopy and to test for contamination. PCA and cluster analysis were conducted using the freely available software PAST (Hammer *et al.* 2001).

RESULTS

Most examples of dinosaur skin are preserved as impressions (e.g. Martill 1991; Kellner 1996; Murphy *et al.* 2002; Schweitzer 2011). In contrast, the integumentary structure in YPMPU 016969 shows internal structure and is preserved as three-dimensional mineralized tissue (Fig. 1A, B). The skin shows pebbly and irregular basement scales (*sensu* Bell 2012), which range from 1 to 12 mm in diameter. The upper and lower margins of the preserved skin are evident in thin section because of contrasts in structure and colouration with the adjacent sediment (Fig. 2A). Two layers can be recognized in thin section. The outer layer, 0.1–0.2 mm thick, characterized by fine stratification overlying organic rich, opaque

bodies composed of melanin (Fig. 2C), is interpreted as the epidermis. It shows similarities to the epidermis of extant vertebrates (Alibardi 2003; Prum & Torres 2003, 2004; Allam *et al.* 2016). The inner layer, 2–3 mm thick, is interpreted as the dermis (Fig. 2B–E) based on its position in the skin and its greater thickness. The difference in thickness and arrangement of the epidermis and dermis in YPMMPU 016969 are similar to those observed in other hadrosaurs (Manning *et al.* 2009; Barbi *et al.* 2019; Fig. 2) and reflect the plesiomorphic condition in extant reptiles and birds (Alibardi 2003; Prum & Torres 2003; Allam *et al.* 2016). EDS reveals a higher concentration of calcium in the dermis than in the adjacent sediment, indicating early cementation. There is no evidence that the innermost layer, the subcutis, is preserved. The total preserved thickness of the skin averages *c.* 3 mm and reaches a maximum (3.3 mm) in association with the largest scales.

Concentrations of Si, Al, Na, O and Ca (Fig. 3) indicate the presence of clay minerals and carbonates, which obscure details of the microstructure of the dermis (Fig. 3; Fabbri *et al.* 2019, fig. S2). Following decay, some sediment infiltrated the space separating the skin from the underlying rib. The concentration of Ca in the dermis (Fig. 3; Fabbri *et al.* 2019, fig. S2) indicates early carbonate cementation as reported by Manning *et al.* (2009) in their investigation of similarly preserved hadrosaur skin. A high concentration of carbon is present only in the opaque layer within the epidermis (Fig. 3; Fabbri *et al.* 2019, fig. S2). Raman microspectroscopy shows that this carbon-rich layer is composed of diagenetically altered eumelanin in a degraded extracellular proteinaceous matrix (Fig. 4A–D; Fabbri *et al.* 2019, fig. S3). No traces of sulfur or functional groups associated with the benzothiazine monomer characteristic of phaeomelanin were found (Figs 3–4; Fabbri *et al.*

2019, fig. S2). SEM of the proposed epidermal layer revealed an amorphous material, showing no evidence of melanosome morphology (Fig. 2E). Vascular networks, fragments of formerly proteinaceous matrix and globular structures resembling cells were extracted by decalcification of epidermis, dermis and bone (Fig. 4B–D; Fabbri *et al.* 2019, fig. S3). The morphologically best-preserved residues (extracellular matrix, degraded blood vessels and osteocytes; Fig. 4B; Fabbri *et al.* 2019, fig. S3) were obtained from the cortical rib bone. Patches of avascular, originally proteinaceous extracellular matrix embedding small, dark-brown eumelanin-rich granules from 0.2 to 0.5 mm in diameter were recovered from the epidermis. Vascular networks invading homogenous, originally proteinaceous extracellular matrix were found exclusively in the dermis (Fig. 4C; Fabbri *et al.* 2019, fig. S3) providing further evidence for a distinct stratification of the skin. Cellular structures were identified based on their globular shape, size and filipodia (Fig. 4B–E). Raman spectroscopy confirmed their organic nature and detected oxidative protein transformation products but no amino acid residues. Acidic decalcification of sediment between the rib and the skin, and that covering the outer surface of the skin, yielded no cellular or matrix organic residues, only clay minerals (Fig. 4E). The organics obtained through decalcification and the C-rich amorphous bodies in the epidermis are endogenous to the skin (Fig. 4F).

We performed PCA to discriminate between the different sample sets and to test for possible recent contamination (Fig. 4F). PC1 mainly represents C=C units, and C-N frameworks prominent in advanced glycoxidation endproducts (AGEs) and advanced lipoxidation endproducts (ALEs) (bands 1400–1430/cm, 1434–1440/cm), while PC2 represents O=C-N units (amides: peptide bonds and/or diagenetically formed lactams) and N-heterocycles (bands 1230–1250/cm, 1340–1352/cm). The dermis and bone spectra nest within the group of fossilized organic material, whereas the sediment falls within the control samples. Three distinct compound clusters were recovered (Fig. 4F): the fossil organic compounds including those from the dermis and bone of YPMMPU 016969; the recent samples; and the blanks and potential contaminants including sediment from YPMMPU 016969, thereby confirming these associations.

DISCUSSION

Our interpretation of three-dimensionally preserved skin in YPMMPU 016969 is based on the structural preservation of layers, diagnostic cellular and organic matrix morphologies extracted from them, and chemical characterization of those layers and organic structures

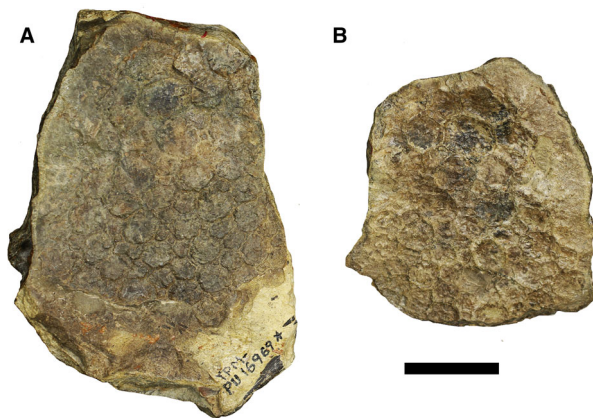


FIG. 1. The skin preserved in YPMMPU 016969. A, three-dimensional skin. B, counterpart. Scale bar represents 20 mm.

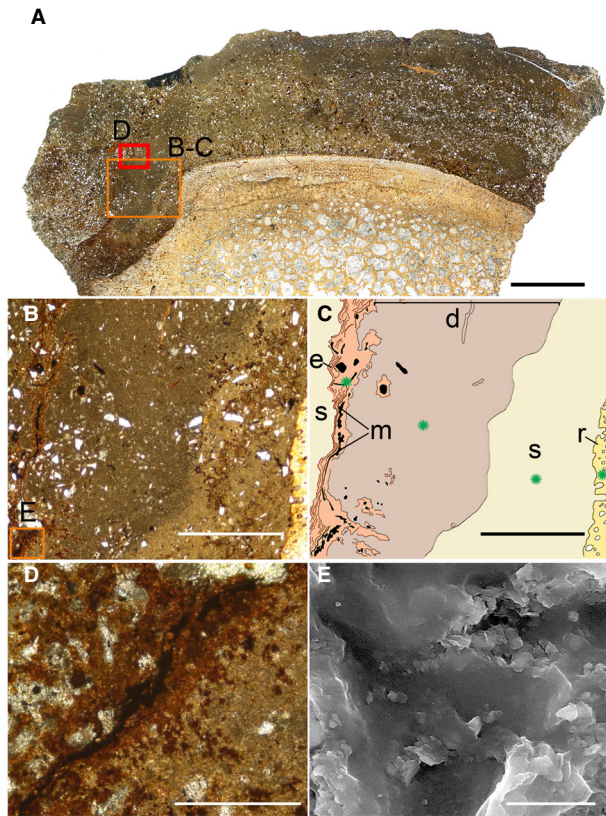


FIG. 2. Morphological characterization of the skin preserved in YPMPU 016969. A, thin section of mineralized skin and underlying bone; red rectangle indicates the area investigated with EDS chemical mapping (Fig. 3). B–C, skin with interpretative drawing showing its structure; we interpret the outermost layer of the skin as the epidermis (e; orange in C), based on its outer position and presence of layering; the dermis is here interpreted as the greyish, thicker portion of the skin lying below and in contact with the epidermis; the difference in thickness between epidermis and dermis, and their relative positions, reflects the condition observed in modern reptiles (including birds) and is consistent with integumentary structures reported in other hadrosaurs (Manning *et al.* 2009); sediment is present on top of the skin and between the rib and the integument; green stars indicate where the samples of epidermis, dermis, sediment and rib (from left to right) were extracted for decalcification and Raman spectroscopy analysis. D, detail of the opaque layer in the outermost portion of the dermis. E, SEM of the carbon-rich layer within the dermis; melanosome morphology is not preserved. *Abbreviations:* d, dermis; e, epidermis; m, melanin; r, rib; s, sediment. Scale bars represent: 5 mm (A); 2 mm (B–C); 1 mm (D); 1 μ m (E).

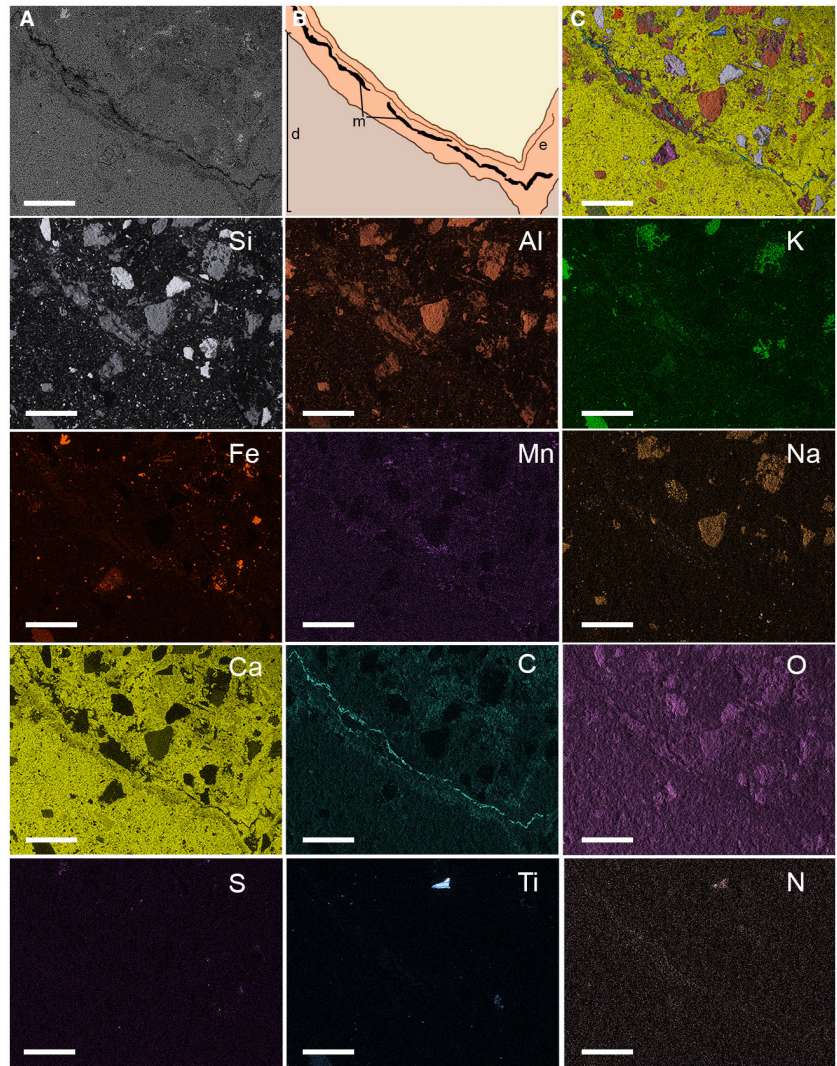
obtained by decalcifying them. The mineral composition of the dermis is interpreted as a result of precipitation of authigenic minerals, including calcite, following decay of more labile organic molecular components, e.g. proteins and lipids.

The thin section shows that the dark C-rich layer occupies a consistent position within the outermost zone of the epidermis (Fig. 2). EDS chemical mapping revealed a high concentration of carbon coincident with the opaque layer observed in the thin section (Fig. 3). The Raman spectra indicate that this layer is organic in nature and incorporates eumelanin (Fig. 4). This is the first report of a specimen preserving a eumelanin-rich layer within three-dimensionally preserved dinosaur skin. Melanosome morphology is not preserved, but its loss has also been noted in some fossil feathers (Li *et al.* 2010; Carney *et al.* 2012), in the skin of the Cretaceous nodosaur *Borealopelta* (Brown *et al.* 2017) and in experimental studies (Colleary *et al.* 2015). The presence of eumelanin may imply a grey colour, similar to that of the skin of living rhinoceros and elephants. However, we did not detect chromatophores and there is insufficient evidence to determine the possible influence of skin structure, variables that are known to have an important role in determining the colouration of the skin in reptiles (Kuriyama *et al.* 2006; Lindgren *et al.* 2014). Therefore, we cannot reconstruct the original colouration of the hadrosaur definitively.

The relative thickness of the epidermis and dermis in YPMPU 016969 resembles that in birds more closely than that of reptiles (Manning *et al.* 2009). These layers differ in structure and composition in birds (Baumel 1993; Alibardi 2003; Prum & Torres 2003) and may vary in their susceptibility to decay (Hovhannisyan *et al.* 2009; Onuh *et al.* 2014) but we did not detect differences in their mineralogical and elemental composition in the fossil. The similarity in skin structure and thickness between hadrosaurs and birds, a resemblance also noted in other studies (e.g. Padian *et al.* 2001; Manning *et al.* 2009; Erickson *et al.* 2017), may reflect physiological similarities.

Previous studies have suggested that the greater prevalence of skin preservation in hadrosaurs compared to other dinosaurs reflects the greater thickness of their integument or a higher content of β -keratin (Lyson & Longrich 2011; Bell 2012; Davis 2014). It is unlikely that the thickness of the skin in YPMPU 016969 changed significantly during fossilization apart from the loss of the subcutis. Proteinaceous remains rely on early diagenetic mineralization for their preservation (Wiemann *et al.* 2018) reducing the likelihood of shrinkage and dehydration. The thickness of the epidermis is similar to that in the hadrosaur from North Dakota (Manning *et al.* 2009) and a more recently described specimen from Alberta, Canada (Barbi *et al.* 2019); the dermis is slightly thinner. It is unlikely that the thinner dermis in YPMPU 016969 reflects an earlier ontogenetic stage; the ontogenetic stage (see Fabbri *et al.* 2019) is comparable to that of the previously investigated

FIG. 3. EDS chemical maps. A–B, SEM image of the area of the skin investigated with EDS; B, interpretative drawing. C, global map showing element distribution obtained by overlap of the single element maps obtained with EDS chemical mapping. *Maps:* Al, aluminium; C, carbon; Ca, calcium; Fe, iron; K, potassium; Mn, manganese; N, nitrogen; Na, sodium; O, oxygen; S, sulfur; Si, silicon; Ti, titanium. Scale bars represent 0.2 mm.



hadrosaur specimens. This difference may reflect the different regions of the body investigated (the ribcage vs limbs, neck and tail) and is consistent with previous observations on mammals (Matoltsy 1986; Shadwick *et al.* 1992). Our new results show that mineralized skin in hadrosaurs (taking the loss of the subcutis into account) is thinner than that of extant mammals of similar size, such as rhinoceros (15–25 mm) and elephants (10–15 mm) (Matoltsy 1986; Shadwick *et al.* 1992) and also thinner than that of the small ceratopsian *Psittacosaurus*, which was estimated at *c.* 10 mm (Lingham-Soliar 2008). If thickness were a key factor determining the preservation of skin, only particular regions of the body would be represented. Thus, thickness does not appear to be a reliable explanation for the high frequency of skin preservation in hadrosaurs. Alternatively, its preservation may reflect the unusual layering and microstructure of hadrosaur skin.

CONCLUSIONS

Our results demonstrate that mineralized dinosaur integument can yield fragments of proteinaceous soft tissues. The structure of hadrosaur skin suggests that these dinosaurs may have had a similar physiology to Aves. Colouration of the skin may reflect ecological parallels to large-bodied extant mammals. The composition and structure of the skin may be important factors in explaining the abundance of hadrosaur skin in the fossil record.

Acknowledgements. We are grateful to Matthew Davis, Guillermo Navalón, Lidya Tarhan, Jakob Vinther and Kanani Lee for discussion and access to facilities at the Yale Institute for the Preservation of Cultural Heritage (IPCH). Support was provided by the IPCH and the Yale Peabody Museum of Natural History. We also thank two anonymous referees and Sally Thomas for helpful comments on earlier drafts of this paper.

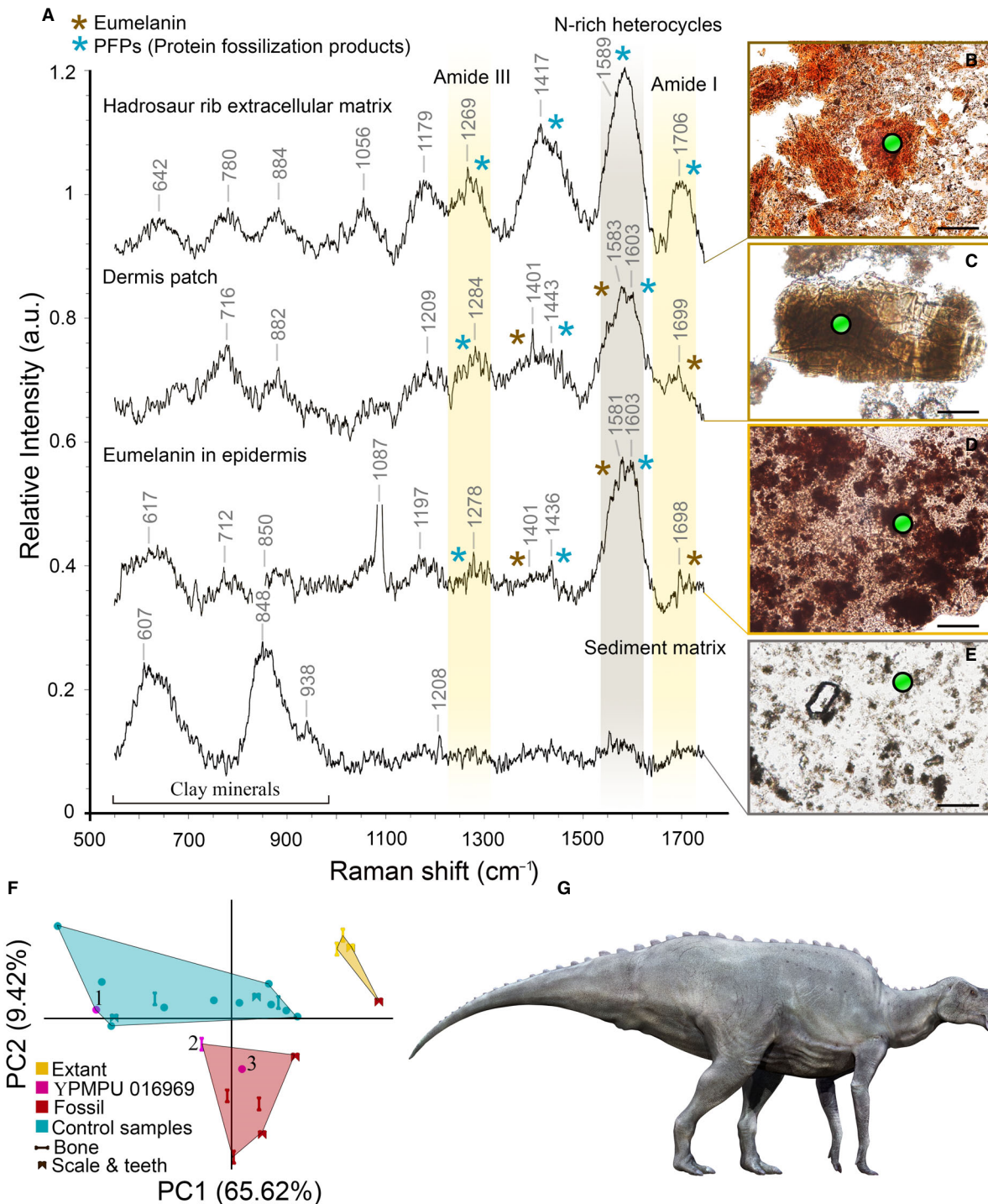


FIG. 4. Raman microspectroscopy of the soft tissues from YPMPU 016969, PCA analysis of samples, and reconstruction of the hadrosaur. **A**, spectra depicting the chemical composition of the soft tissues obtained from decalcification of bone, skin layers and sediment. **B**, soft tissues from the rib. **C**, vascular canals from the epidermis. **D**, formerly proteinaceous matrix from the dermis with embedded melanocytes. **E**, clay minerals from the sediment. Green dots indicate where samples were analysed with Raman microspectroscopy. **F**, PCA plot modified after Wiemann *et al.* (2018) showing the position of samples from the sediment, bone and epidermis in YPMPU 016969 (1, sediment; 2, organics from the rib; 3, organics from the epidermis; blue, blanks; red, fossil samples; yellow, modern samples). **G**, reconstruction of YPMPU 016969 based on the skeleton, skin patches and colouration reported in this study; peak 1603 is used in this study as indole function for eumelanin, following Galván *et al.* (2013) and Huang *et al.* (2004). Scale bars represent 0.3 mm (B–E).

Author contributions. MF designed the project, interpreted morphology, and performed SEM and EDS analyses, JW prepared and analysed samples with Raman spectroscopy, FM reconstructed the animal, MF interpreted the data and wrote the paper with input from all the other authors.

DATA ARCHIVING STATEMENT

Data for this study are available in the Dryad Digital Repository: <https://doi.org/10.5061/dryad.3bk3j9kfl>

Editor. Robert Sansom

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